AUTOSAR High-Level Design (HLD) Specification

Document ID: HLD-EBC-2026-V1.3 | System: Electronic Brake Control (EBC) ECU

Section 1. System Overview & Architecture Goals

This High-Level Design document specifies the AUTOSAR Classic Software Architecture for the Electronic Brake Control (EBC) ECU. The primary functional safety requirement is to acquire wheel speed signals, evaluate vehicle deceleration dynamics, execute Anti-Lock Braking System (ABS) algorithms, and command hydraulic actuator valves in real time.

Section 2. Software Component (SW-C) Definitions

SW-C Name	Component Type	Description
WheelSpeedSensor_SWC	Sensor/Actuator	Acquires raw pulse frequencies from 4 wheel speed sensors and converts them to speed.
BrakeController_SWC	Application	Executes slip ratio estimation and ABS brake pressure modulation algorithms.
ActuatorValves_SWC	Sensor/Actuator	Drives PWM hardware interfaces to hydraulic solenoid valves.
DiagnosticManager_SWC	Service Component	Monitors DTC event codes and reports safety bus faults.
HillHoldAssist_SWC	Application	Holds brake pressure on inclines until drive-off.

Section 3. AUTOSAR Interfaces & Port Definitions

The components communicate using AUTOSAR Sender-Receiver and Client-Server interfaces. WheelSpeedSensor_SWC provides P_WheelSpeed using WheelSpeed_If. BrakeController_SWC requires R_WheelSpeed and provides P_BrakeRequest using BrakeRequest_If. ActuatorValves_SWC requires R_ValveCmd using ValveControl_If.

Port Name	Direction	SW-C Owner	Interface Reference
P_WheelSpeed	Provided	WheelSpeedSensor_SWC	WheelSpeed_If
R_WheelSpeed	Required	BrakeController_SWC	WheelSpeed_If
P_BrakeRequest	Provided	BrakeController_SWC	BrakeRequest_If
P_ValveCmd	Provided	BrakeController_SWC	ValveControl_If
R_ValveCmd	Required	ActuatorValves_SWC	ValveControl_If
R_VehicleSpeed	Required	HillHoldAssist_SWC	VehicleSpeed_If

Section 4. Signal Dictionary & CAN Bus Mapping

Signal Name	Data Type	Unit	Interface Ref	Description
FL_WheelSpeed	float32	km/h	WheelSpeed_If	Front Left Wheel velocity
FR_WheelSpeed	float32	km/h	WheelSpeed_If	Front Right Wheel velocity
RL_WheelSpeed	float32	km/h	WheelSpeed_If	Rear Left Wheel velocity
RR_WheelSpeed	float32	km/h	WheelSpeed_If	Rear Right Wheel velocity
BrakePressureCmd	float32	bar	BrakeRequest_If	Hydraulic pressure command
ABS_ActiveFlag	boolean	-	BrakeRequest_If	Indicates active ABS slip control
VehicleSpeed	float32	km/h	VehicleSpeed_If	Reference vehicle speed

Section 5. Component Dependencies & Functional Flows

Component Dependencies: - WheelSpeedSensor_SWC -> BrakeController_SWC (via WheelSpeed_If) - BrakeController_SWC -> ActuatorValves_SWC (via ValveControl_If) - BrakeController_SWC -> DiagnosticManager_SWC (via DiagnosticReport_If) - BrakeController_SWC -> ParkBrake_SWC (via ParkBrake_If)

Execution Flow: Flow: WheelSpeedSensor_SWC -> BrakeController_SWC -> ActuatorValves_SWC -> DiagnosticManager_SWC

Description of Flow: The physical sensor generates pulses acquired by WheelSpeedSensor_SWC. The speed data is transmitted over WheelSpeed_If to BrakeController_SWC, which determines slip control. Pressure actuation commands are sent over ValveControl_If to ActuatorValves_SWC, and diagnostic status is logged by DiagnosticManager_SWC.